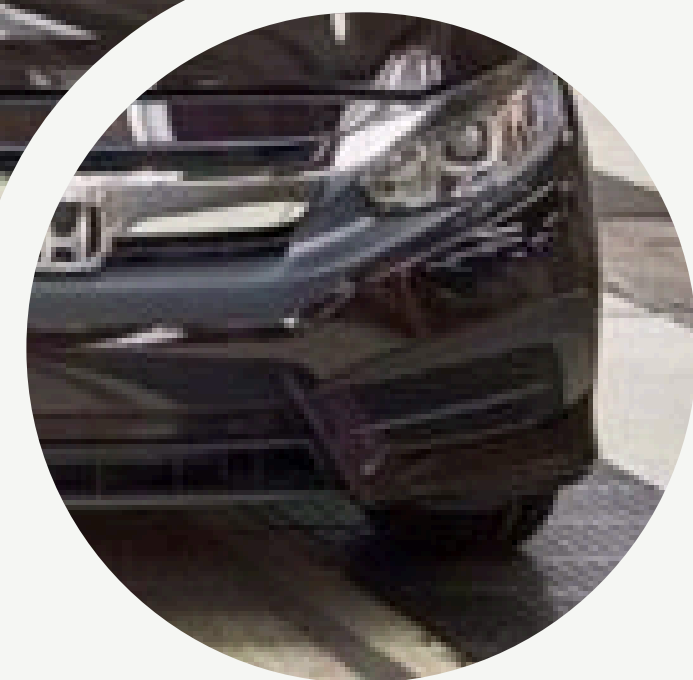
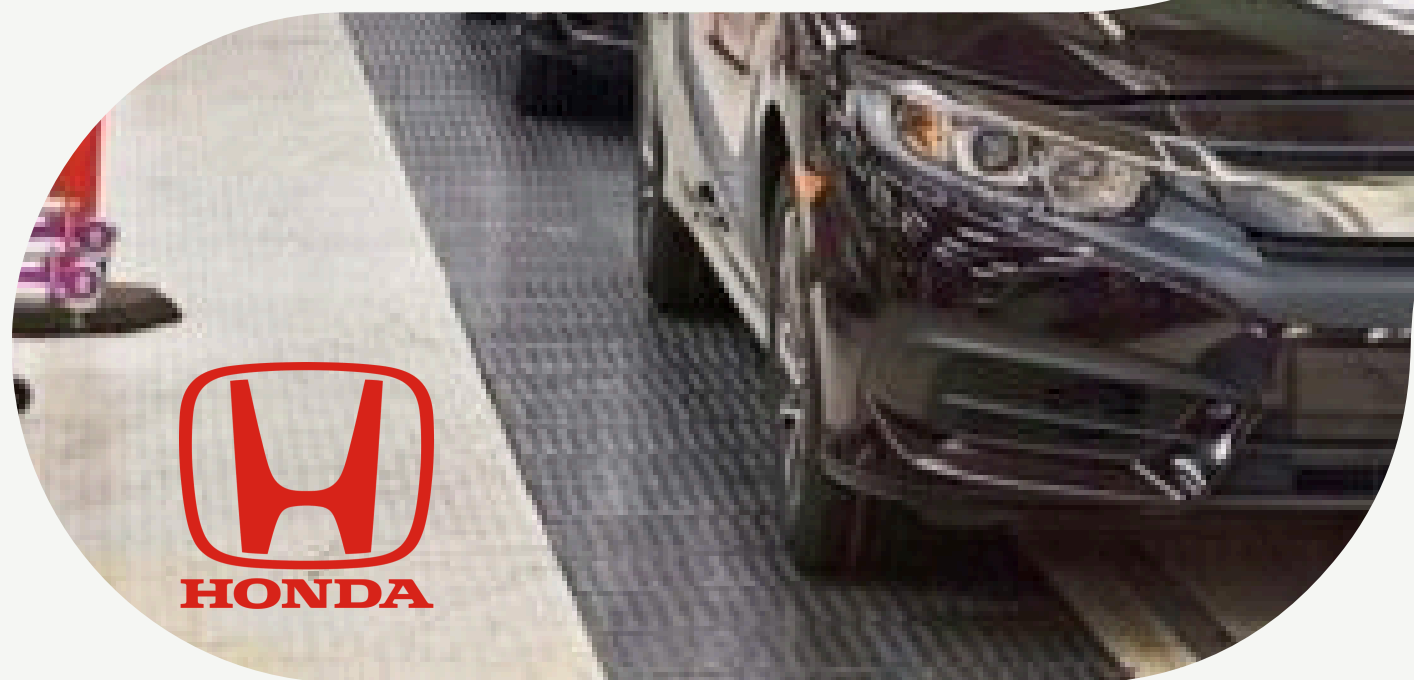
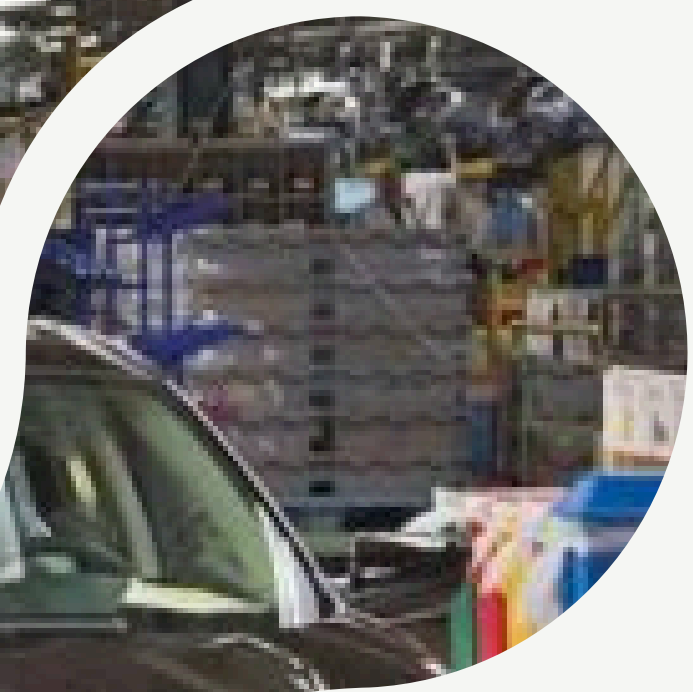
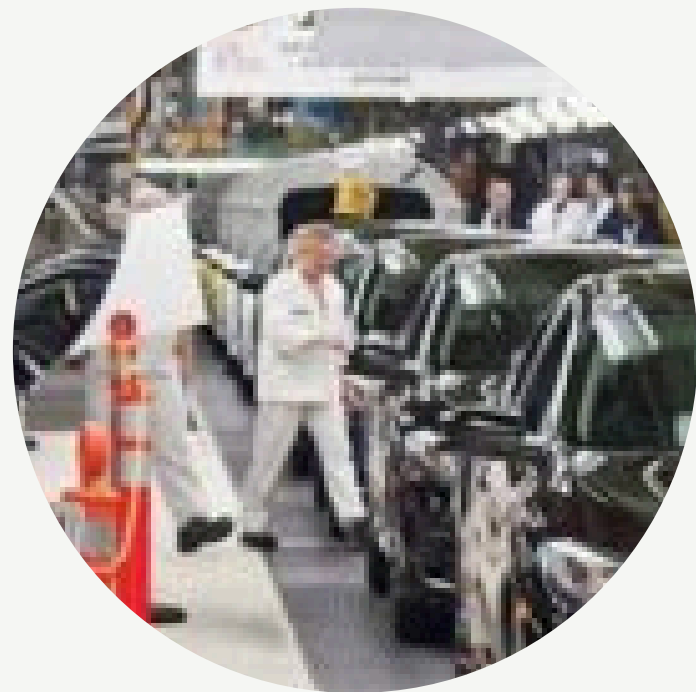


V6



Naz Mahdian
Remy Post
Vaibhav Narula
Ken Hein
David Quinn
Susan Soto

**Why do we predict the
Machine Failure, but
NOT the human
conditions that cause
it?**



PREDICT

Because we need signals that we can detect today that help predict a failure tomorrow

Man

Fatigue Devices that measure the probability of human error. Eye blink rate, slower reaction time, reduced motions.

Weekly seminars

Mental Health Seminars: Encouraging Employees to Open Up
Studies indicate that roughly 34% to 55% of employees do not speak up about issues or ideas at work,

Material problems

We need engineers to ask why did the system allow the mistake

using acoustic sensors to hear tiny crack activity that could jeopardize the production

Improper Methods

Would be teams being reactive rather than being proactive



PREVENT



Human

Confidence

Material problems

Maintenance

**Human-Robot
synergy**

improper Methods

No need to ask for WHO

FIND OUT THE WHY

Cybersecurity

Look for a pattern



DATA

AI As a complement

Implement with HUMAN expertise





FAST RECOVERY

Human

Empower employees on
self diagnosing the
problem

Learning through failure

Material problems

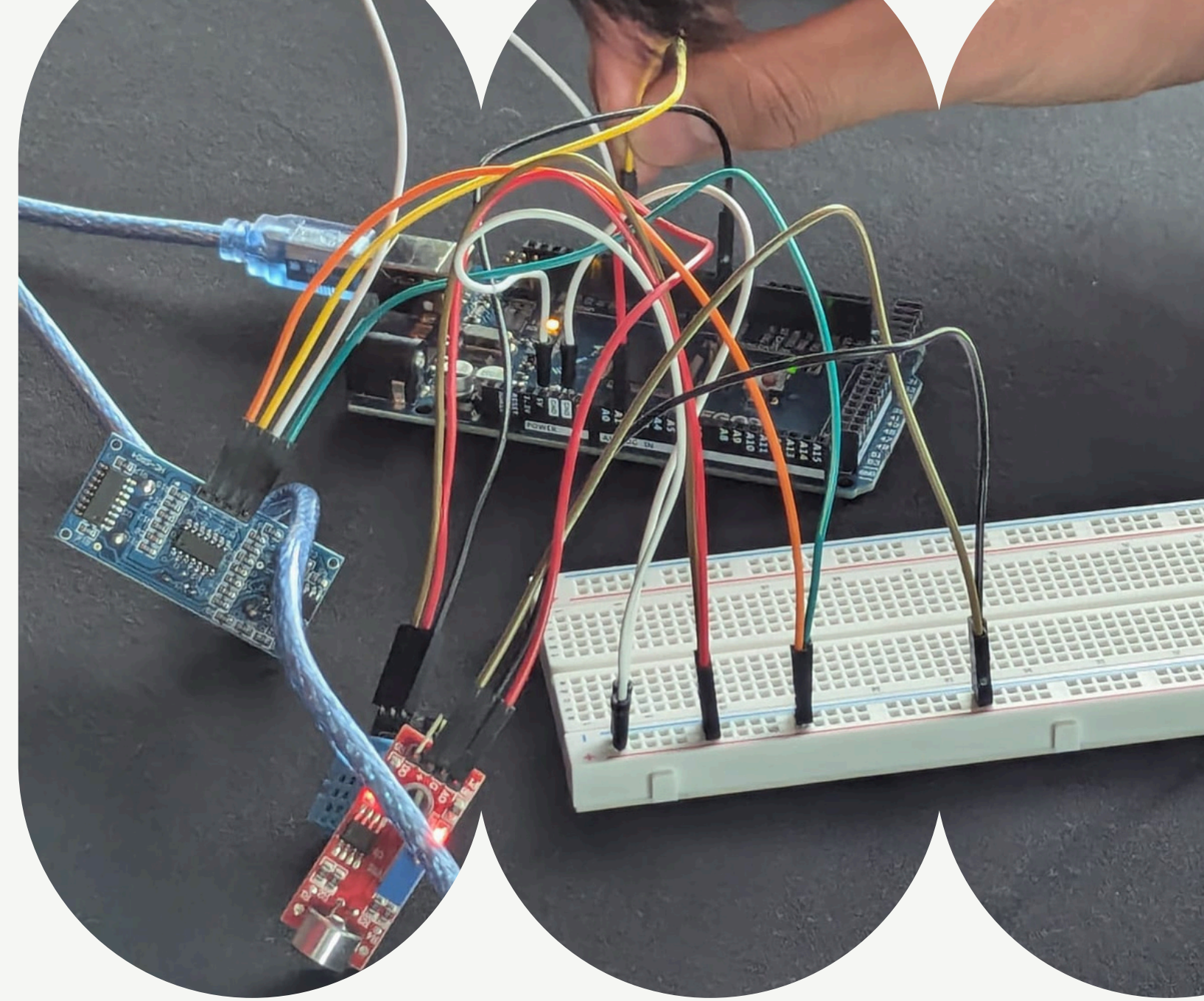
AI intelligence to guide
the employees to
trouble shoot the
problem

improper Methods

Hardware Issues &
Environmental issues
and power outages
-Work Collaboratively



TECHNICAL INNOVATION



PREDICTION

PREVENTION

RECOVERY





Prediction & Prevention

OVERALL CONDITION

FAIR

VIBRATION ANOMALY — MIDDLE ARM

Abnormal vibration detected at middle arm — *Inspect middle arm for loose components or bearing wear*

CRITICAL

VIBRATION — MIDDLE ARM

Critical vibration at MIDDLE ARM (delta: 1023) — stop machine immediately

15:10:48

CRITICAL

VIBRATION — TOOL END

Critical vibration at TOOL END (delta: 795) — stop machine immediately

15:10:48

NOTICE

TEMPERATURE — TOOL END

Temp rising at TOOL END: 37.6°C (+11.5°C above baseline)

15:10:48

WARNING

HUMIDITY — TOOL END

High humidity at TOOL END: 26.6% (+19.4% above baseline)

15:10:48

CONDITION REPORT

TEMPERATURE

TEMP — BASE

EXCELLENT

TEMP — JOINT

EXCELLENT

TEMP — TOOL

FAIR

HUMIDITY

HUMID — BASE

EXCELLENT

HUMID — JOINT

EXCELLENT

HUMID — TOOL

FAIR

VIBRATION

VIB — BASE

EXCELLENT

VIB — MID

CRITICAL

LIVE READINGS

TEMPERATURE

BASE

23.8°C

+0.3 from base

JOINT

40.7°C

+0.2 from base

TOOL

37.6°C

+11.5 from base

HUMIDITY

BASE

45%

-1.0 from base

JOINT

22%

0.0 from base

TOOL

26.6%

-19.4 from base

VIBRATION

BASE

0

0.0 from base

MID

0

-1023.0 from base

TOOL

121

-795.0 from base

CONDITION REPORT

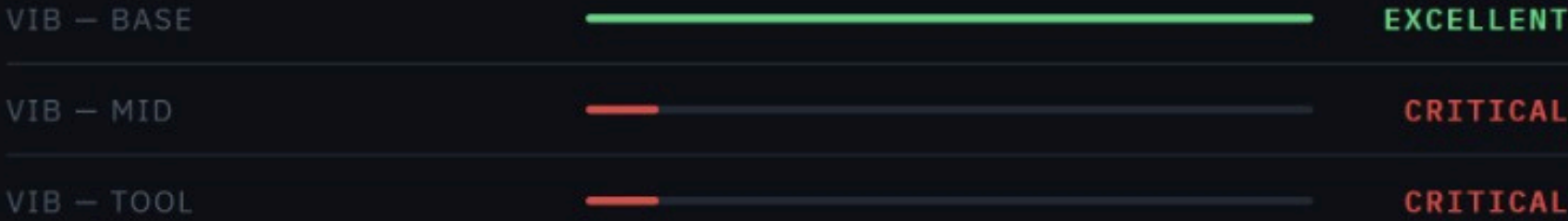
TEMPERATURE



HUMIDITY



VIBRATION



OTHER

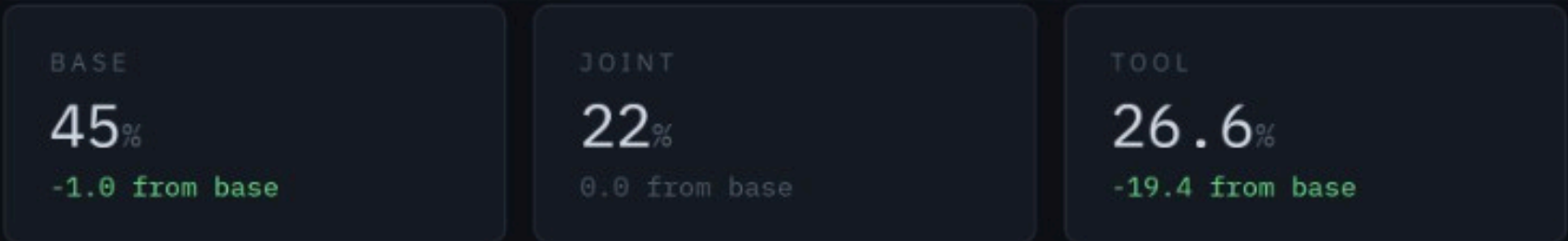


LIVE READINGS

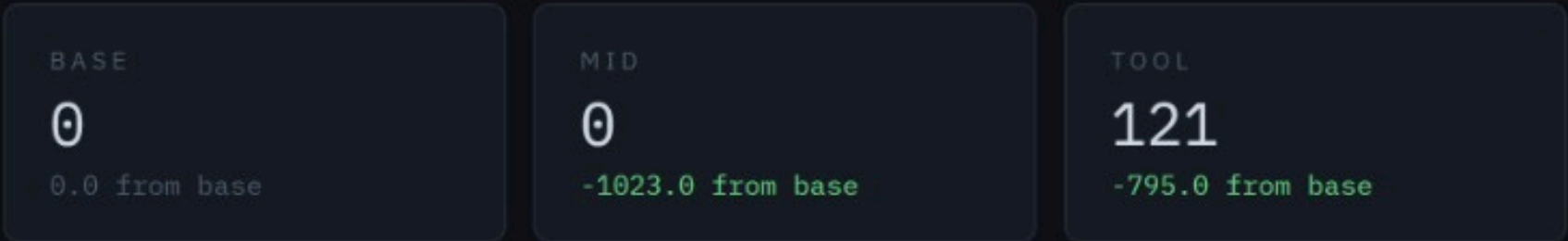
TEMPERATURE



HUMIDITY



VIBRATION



OTHER



TEMPERATURE TREND — BASE SENSOR



PARTS BREAKDOWN

FANUC R-2000iA/165F

Axes: 6 | Payload: 165 kg | Reach: 2655 mm

● Sensor feed offline

J1 — Waist Rotation ●

▸ Aluminum alloy housing
↓ 50.7°C 📊 0.15g ⚡ 40%

J2 — Shoulder (Lower Arm) ●

▸ High-strength steel
↓ 50.0°C 📊 0.04g ⚡ 68%

J3 — Elbow (Upper Arm) ●

▸ Aluminum alloy
↓ 51.7°C 📊 0.04g ⚡ 37%

J4 — Wrist Roll ●

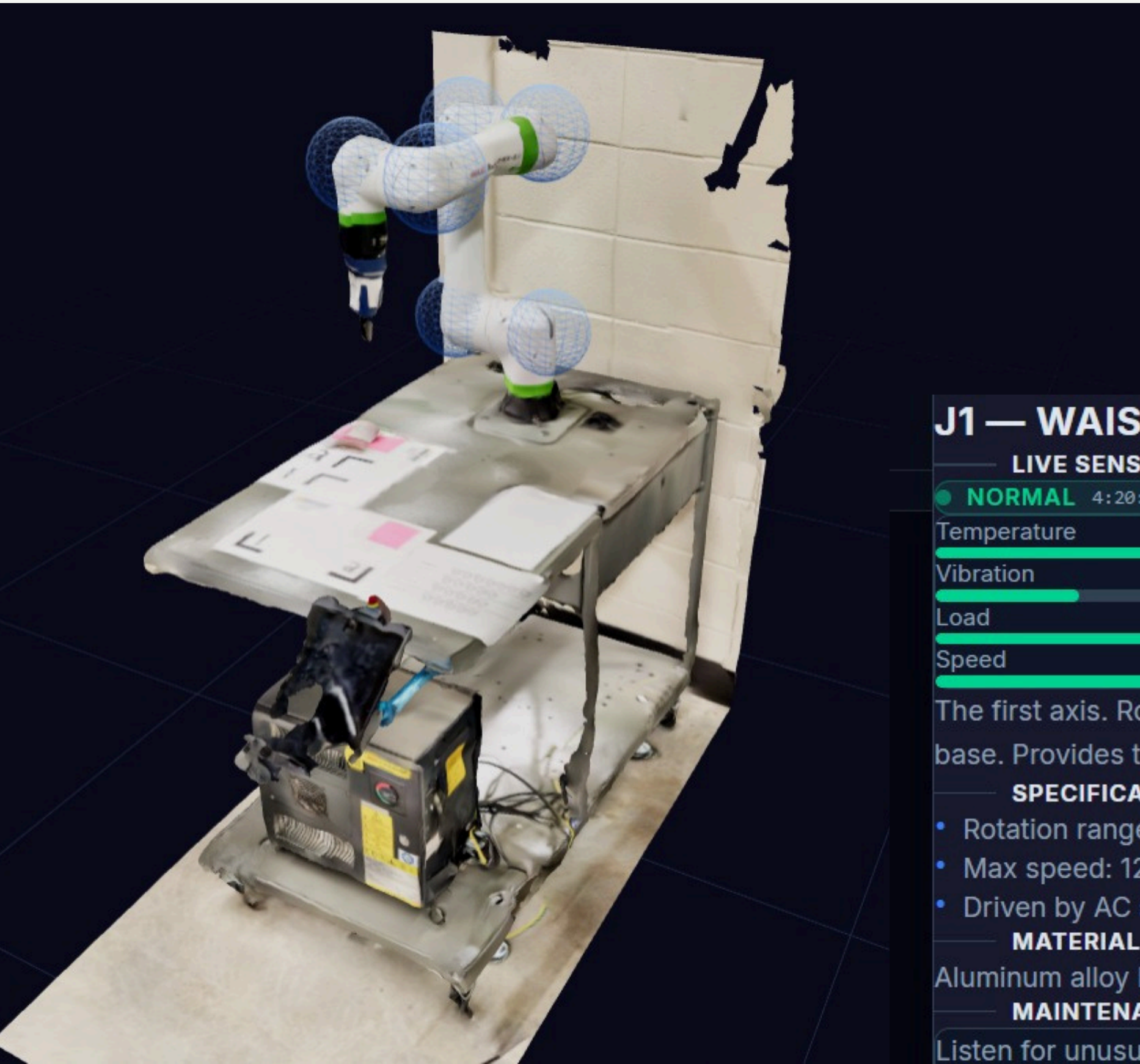
▸ Aluminum alloy
↓ 36.4°C 📊 0.22g ⚡ 33%

J5 — Wrist Pitch ●

▸ Aluminum alloy
↓ 45.7°C 📊 0.20g ⚡ 69%

J6 — Wrist Yaw (Tool Flange) ●

▸ Steel flange, aluminum housing
↓ 51.3°C 📊 0.10g ⚡ 43%



! ERROR LOG

Live & recent exclamation alerts

Live Errors

No live errors.

Recent Errors

No recent errors.

J1 — WAIST ROTATION

LIVE SENSOR DATA



NORMAL 4:20:43 PM

Temperature 42.9 °C

Vibration 0.2 g

Load 38.6 %

Speed 101.2 °/s

The first axis. Rotates the entire arm left and right around the base. Provides the robot's full rotational workspace.

SPECIFICATIONS

- Rotation range: $\pm 360^\circ$
- Max speed: $120^\circ/\text{sec}$
- Driven by AC servo motor + harmonic drive

MATERIAL

Aluminum alloy housing

MAINTENANCE NOTES

Listen for unusual grinding sounds during rotation. Monitor motor temperature. Grease harmonic drive per schedule.



Fast Recovery

Learnings

- Had to transfer raw data from one device to another using IP transfer.
- Learnt that we won't know which joint will fail, we improvise using 3 putting at -prioritizing major failure points

DEMO TIME - MACHINE ISSUES



HUMAN INNOVATION

Keyword

Incentivization structures

- Increase vacation time
- Company Discounts
- Money/ Promotion





Thank you.

Questions?

